

Class 29 More Functions

Goals

1. Reinforce understanding of definitions with examples.
2. Practice proofs involving composition of functions.

Review definitions

f is a function means $a = b$ implies $f(a) = f(b)$ for all a and b in the domain.

f is one-to-one means $f(a) = f(b)$ implies $a = b$. Or the contrapositive $a \neq b$ implies $f(a) \neq f(b)$.

f is onto means for every y in the codomain there is some x in the domain such that $f(x) = y$.

f is a one-to-one correspondence means f is both one-to-one and onto (bijection).

Two functions, f and g , are equal if they have the same domain and $f(x) = g(x)$ for every x in the domain.

Two functions, f and g , are inverses of each other means $g(f(x)) = x$ for all x in the domain of f , and $f(g(y)) = y$ for all y in the range of f .

Examples, Problems, and Proofs

1. Define a mapping $g : M_{2,2} \rightarrow \mathbb{R}$ by $g \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc$

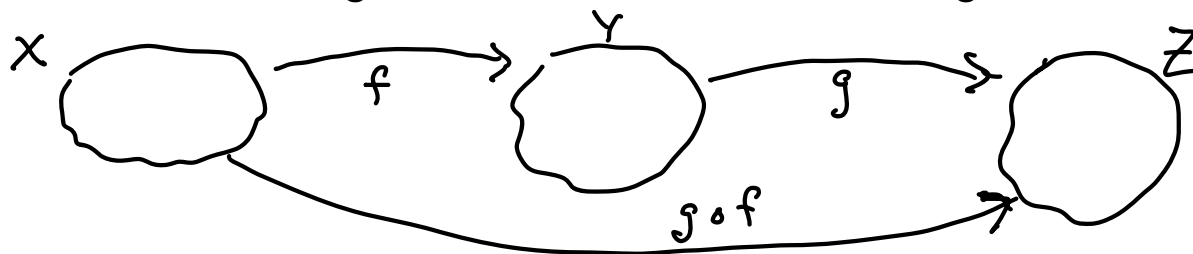
Prove g is not one-to-one but does map onto the set **R**.

2. Define a mapping $f : \mathbb{Q} \rightarrow \mathbb{Z}$ by $f \left(\frac{a}{b} \right) = b$

Prove f is **not** a function. How could you modify the definition to make it a function?

Prove f is **not** one-to-one and **not** onto.

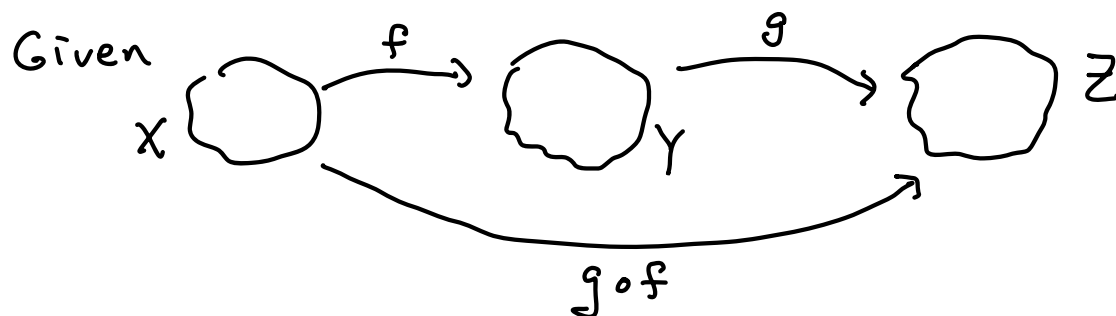
In problems 3 and 4, f and g are functions. $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.



3. If $g \circ f$ is onto, then g is onto.

4. If $g \circ f$ is onto, show that f may or may not be onto.

Summarize: How do the properties of one-to-one and onto work with composition of functions?



If both f and g are onto, then

If $g \circ f$ is onto, then

If both f and g are 1-1, then

If $g \circ f$ is one-to-one, then